Remy3Design

BIMAP

Evidence-First BIM Quality Intelligence

Technical Whitepaper

Integrated audit focus

Revit Family Audit + BIM Deliverable QA + Combined Audit

Prepared for Remy3Design (R3D)

Powered by SLAI v2.3

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Technical architecture and methodology whitepaper

Document Control

This whitepaper defines the BIMAP operating model, technical architecture, evidence methodology, audit scopes, governance model, deliverables, benefits, validation requirements, and staged integration path. It is a technical and methodological document rather than a certification statement or contractual guarantee.

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Scope boundary

BIMAP provides automated pre-review and evidence-linked BIM quality analysis. It does not constitute statutory certification, building-code approval, clash-detection replacement, structural or MEP engineering verification, or a substitute for professional BIM management. Deterministic checks, evidence-linked inferences, unknown conditions, and manual-review requirements are explicitly distinguished.

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1. Executive Summary

Chapter purpose
Define BIMAP, its evidence-first operating model, the relationship between deterministic BIM checks and SLAI-governed reasoning, and the principal value delivered to AEC users.

1.1 Platform definition

BIMAP is an independent BIM quality-intelligence service for Revit families and BIM project deliverables. Its purpose is to convert heterogeneous BIM evidence into structured, traceable findings that state what was assessed, what was observed, which expectation or rule was applied, how certain the assessment is, and what action can be taken next. The architecture is derived from the R3D BIM Audit Platform implementation model and is designed to operate as a commercial service around SLAI v2.3 rather than exposing the SLAI runtime directly to public web traffic (Remy3Design, 2026).

The platform is organized around three audit scopes: Family Audit, BIM QA, and Combined Audit. Family Audit operates on Revit-family and library evidence; BIM QA operates on project information requirements and deliverable evidence; Combined Audit links the two scopes where a family-level data-quality condition is relevant to project-level information quality. The three scopes share ingestion, normalization, evidence identifiers, rule definitions, governance, report generation, and delivery infrastructure.

1.2 Core analytical principle

BIMAP applies a rule-first, reasoning-second model. Deterministic checks establish directly measurable or comparable conditions before contextual interpretation occurs. Contextual reasoning is then constrained to grounded evidence and existing rule outputs for activities such as cross-document relationship analysis, prioritization, explanation, unresolved-condition handling, and report synthesis. This separation supports reproducibility and keeps probabilistic inference distinct from direct evidence.

Each material finding therefore carries an automation type, a severity, a confidence value, evidence references, expected and observed states, remediation guidance, and a verification method. Severity describes potential impact if a finding is valid; confidence describes certainty that the finding is correctly supported. These two dimensions remain independent throughout analysis and reporting.

1.3 Service boundary

BIMAP is defined as decision support and automated pre-review. It is not a statutory certification body, building-code authority, geometric clash-detection engine, structural or MEP engineering verifier, or substitute for professional BIM management. Unsupported conditions remain visible as unknown, not assessed, or manual-review-required rather than being converted into unsupported compliance claims (Remy3Design, 2026).

This boundary is central to the whitepaper: the system is designed to improve the structure, traceability, consistency, and prioritization of BIM quality review while preserving explicit limits on what can be established from available evidence.

2. Context and Problem Definition

Chapter purpose
Establish the information-quality problem addressed by BIMAP and distinguish BIM quality intelligence from statutory compliance, geometric clash detection, and professional design verification.

2.1 Information quality in BIM workflows

Building information modelling depends on the reliable creation, exchange, versioning, and organization of information across project and asset life cycles. ISO 19650 frames information management around these coordinated processes, while BIM practice literature consistently identifies structured information, model standards, and controlled exchanges as necessary conditions for dependable digital delivery (International Organization for Standardization, 2018; Sacks et al., 2018).

In practice, quality problems often occur across different information layers. A Revit family may contain inconsistent parameter definitions, naming, type/instance decisions, formulas, materials, connectors, or documentation. A project deliverable may contain missing required information, inconsistent identifiers, revision conflicts, register mismatches, incomplete provenance, or weak alignment with project information requirements. These defects may remain isolated, or they may propagate from content libraries into schedules and project deliverables.

2.2 Why evidence-first auditing is necessary

Automated BIM checking is not a single computational problem. Rules differ in the type of data they require and in whether they can be converted into deterministic logic. Solihin and Eastman (2015) classify automated BIM rules according to their computational characteristics, illustrating why some requirements can be checked directly while others require interpretation or remain unsuitable for full automation.

BIMAP responds by preserving the distinction between observable evidence, deterministic rule results, contextual inference, and unresolved conditions. A missing parameter in a structured family export may support a deterministic finding. A contradiction between two documents may require cross-source reasoning. A condition not represented in the evidence package cannot be treated as observed merely because it is plausible.

2.3 Design objective

The platform objective is to provide a structured second-review layer that combines heterogeneous evidence, explicit rules, source provenance, prioritized remediation, and governed contextual analysis. The resulting BIM health report is intended to be auditable: each material statement can be traced to the source evidence and rule or requirement that produced it.

This approach differentiates BIMAP from opaque scoring systems. Rather than relying on a single composite “AI score,” the platform can expose evidence coverage, deterministic pass rates, severity distribution, unknown or missing-evidence counts, and confidence profiles. Composite scoring can only be introduced responsibly after empirical calibration against expert assessment.

3. BIMAP Product Model

Chapter purpose
Define the Family Audit, BIM QA, and Combined Audit scopes and explain how they share one evidence pipeline, one finding model, and one reporting contract.

3.1 Family Audit

Family Audit evaluates evidence about Revit families against configured technical and organizational criteria. Initial deterministic domains include identity and category consistency, family/type naming, parameter governance, data types, required parameters, type-versus-instance policy, formulas where exported, standards comparison, and documentation completeness. Material and connector checks depend on the evidence available from a future Revit-native exporter.

The audit output is not limited to a list of failed checks. It contains a family inventory, source manifest, categorized findings, unsupported conditions, remediation actions, and verification methods that indicate how an issue can be closed after correction.

3.2 BIM QA

BIM QA evaluates project deliverables against explicit information requirements. Relevant evidence includes BIM execution plans, exchange or employer information requirements, schedules, drawing and issue registers, model-checker exports, IFC or IFC-derived reports, spreadsheets, PDFs, and selected screenshots. The core analytical artifact is a Requirement-Evidence Matrix that records the requirement, supporting evidence, assessment state, automation type, confidence, impact, and recommended action.

Assessment states include pass, warn, fail, unknown, and not_applicable. Unknown is a first-class state because absence of evidence is not equivalent to non-conformance unless the requirement itself explicitly requires evidence to be present.

3.3 Combined Audit

Combined Audit executes both scopes under one order and one evidence model. Family findings can become corroborating or explanatory project evidence when the same data condition appears in schedules, registers, or other deliverables. Project requirements can also indicate which family defects carry the greatest project impact.

The combined result uses typed relationships such as causes, contributes_to, duplicates, conflicts_with, depends_on, and corroborates. Typed relationships are required because a statistical or narrative association does not automatically establish causation.

3.4 Product comparison

Product	Primary object	Typical evidence	Primary output
Family Audit	Revit family or library evidence	Family exports, schedules, screenshots, family standards	Family health report, findings.json, remediation.csv
BIM QA	Project information deliverables	BEP/EIR, schedules, registers, IFC reports, checker exports	BIM health report, requirement-evidence matrix, remediation.csv
Combined Audit	Families + project evidence	Both evidence groups under one audit context	Integrated risk summary, linked findings, combined remediation plan

4. System Architecture

Chapter purpose
Describe the layered technical architecture that separates the frontend, service layer, audit engine, SLAI integration boundary, persistence, workers, and deterministic reporting.

4.1 Layered service architecture

BIMAP separates the public service layer from the audit engine and from SLAI. The frontend is responsible for product communication, upload interaction, customer status, and delivery. The API and application layers own authentication boundaries, order state, file authorization, payment state, queueing, cancellation, and customer-visible lifecycle. The audit engine owns deterministic evidence ingestion, normalization, rules, and grounded findings. The SLAI boundary receives only normalized, policy-approved analysis inputs (Remy3Design, 2026).

This separation prevents untrusted web requests and raw uploaded files from becoming direct agent inputs. It also preserves the audit engine as a testable BIM-specific subsystem whose deterministic results can be validated independently from contextual reasoning.

Figure 1. BIMAP system architecture

Internet-facing service functions remain separated from SLAI and from deterministic BIM rule execution.

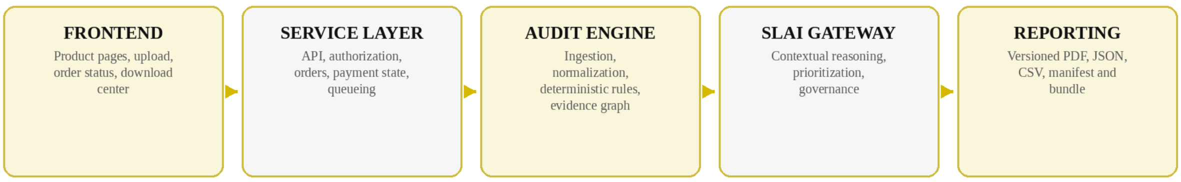

Figure 1. BIMAP layered system architecture.

4.2 Dependency direction

The intended dependency direction follows a layered architecture: frontend traffic enters through the API; API handlers call application commands and queries; application services coordinate domain state, ports, the deterministic audit engine, and the SLAI gateway; workers execute long-running jobs; reporting converts approved results into customer deliverables. Domain objects remain independent of presentation, network, payment, storage, and agent-runtime details.

External technologies are treated as adapters to application ports. The architecture therefore permits storage, database, queue, payment, notification, or malware-scanning implementations to change without modifying core audit rules or domain policy.

4.3 SLAI as an integration dependency

SLAI v2.3 functions as the reasoning and governance engine used by BIMAP. BIMAP remains the owner of product definitions, BIM rules, evidence contracts, order state, customer authorization, report packaging, and commercial lifecycle. SLAI is consumed through a narrow adapter rather than imported throughout the codebase, preserving a clear boundary between the BIM product and the general multi-agent platform (Remy, 2026b).

5. End-to-End Audit Workflow

Chapter purpose
Explain the customer and system lifecycle from product selection and evidence upload through validation, analysis, governance, report generation, delivery, and retention.

5.1 Customer-visible lifecycle

The standard audit lifecycle begins with product selection, followed by evidence upload into a temporary staging area. File-format, integrity, size, signature, and package checks are performed before compute-intensive analysis begins. The service can then complete payment confirmation, release the approved evidence into the processing boundary, enqueue a single analysis job, and expose coarse processing states to the customer.

Customer-visible status does not expose raw agent traces or private internal reasoning. Instead, it reports controlled stages such as uploading, validation, queued, ingesting, analyzing, governance review, packaging, and delivered.

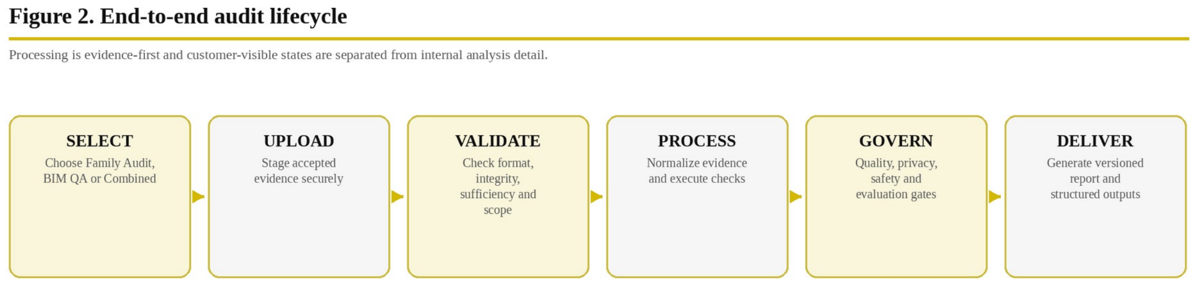

Figure 2. End-to-end BIMAP audit lifecycle.

5.2 Order state machine

Explicit order states provide deterministic workflow control. A reference progression is draft -> uploading -> upload_validated -> payment_pending -> paid -> queued -> ingesting -> analyzing -> governance_review -> packaging -> delivered. Exception states include upload_rejected, payment_failed, analysis_failed, review_required, refunded, cancelled, and expired (Remy3Design, 2026).

Every state transition is validated server-side, timestamped, and recorded so that retries or payment webhooks cannot silently create duplicate jobs. Idempotency is especially important at the transition from payment confirmation to queue submission.

5.3 Long-running processing

Audit analysis is treated as asynchronous work. API processes should remain responsive while workers perform parsing, normalization, rule execution, SLAI orchestration, governance checks, report generation, and retention tasks. This model is compatible with both a simple first implementation and later distributed queue workers.

6. Evidence Acquisition and Normalization

Chapter purpose
Define the evidence-first data strategy, canonical evidence objects, provenance, versioning, and the role of controlled exports before direct Revit processing.

6.1 Controlled evidence package

The initial BIMAP implementation is based on controlled exports rather than assuming that arbitrary RVT or RFA files can be interpreted safely and consistently. Supported evidence can include structured JSON, CSV and XLSX schedules, PDFs, screenshots, issue registers, family metadata, model-checker exports, and project requirement documents. This makes individual fields observable, testable, versionable, and independent of the customer's Revit runtime (Remy3Design, 2026).

IFC and openBIM evidence can be added incrementally. IFC is an open, vendor-neutral data standard published as ISO 16739 and is designed for machine-interpretable exchange across BIM workflows (buildingSMART International, 2024).

6.2 Canonical evidence objects

Raw input formats are normalized into canonical evidence objects before rule execution or contextual reasoning. Each evidence object should preserve evidence_id, source_file_id, source_hash, source_type, logical location such as page, row, element, or path, extractor version, extracted value, and extraction confidence when extraction itself is probabilistic.

Canonicalization prevents independent modules or agents from interpreting the same source differently. It also allows findings to reference stable evidence identifiers instead of fragile filenames or free-text descriptions.

6.3 Provenance and versioning

Provenance is part of the data model rather than an optional report annotation. Source hashes, schema versions, parser or extractor versions, timestamps, and stable identifiers allow the service to identify exactly which evidence produced a report. This supports reproducibility, change comparison, and later Revit-native mapping.

Extraction failures must remain explicit. A missing or unsupported field is represented as an error or unsupported condition in the evidence manifest rather than being silently omitted.

7. Deterministic Audit Engine

Chapter purpose
Explain how BIMAP executes versioned, reproducible checks before contextual reasoning and how rule results become grounded findings, unknown states, and review conditions.

7.1 Rule registry

Every deterministic check is represented by a versioned rule definition. A rule registry records the rule identifier, audit scope, required inputs, rule class, severity policy, deterministic or inferential status, benchmark cases, known limitations, and current version. This structure supports controlled change management and prevents customer-facing behavior from shifting without traceable rule revisions.

Rule classification is important because not all BIM requirements can be reduced to the same computational form. Direct comparisons, presence checks, naming patterns, and type constraints can often be deterministic; contextual requirements may require inference or manual review (Solihin & Eastman, 2015).

7.2 Execution sequence

Rule execution begins only after the evidence package has passed ingestion and normalization. Applicable rules are selected according to the product scope and available evidence. Each rule returns a structured result rather than free-form text. The result identifies the rule, observed state, expected state, source evidence, status, and any deterministic metrics required for later reporting.

A rule may return pass, warn, fail, unknown, or not_applicable. Unknown is used when the evidence required to establish the condition is insufficient. This preserves uncertainty instead of converting absence into false certainty.

Figure 4. Rule-first, reasoning-second analysis

Contextual reasoning operates on grounded evidence and findings rather than replacing deterministic checks.

Figure 4. Rule-first, reasoning-second analysis sequence.

7.3 From rule results to findings

Grounded findings are created from rule results and evidence. A finding records finding_id, scope, rule_id, automation_type, severity, confidence, status, observed_value, expected_value, evidence_refs, explanation, remediation, verification_method, and lifecycle status. The explanation and remediation fields can later be enriched by governed contextual reasoning, but the underlying evidence and deterministic result remain authoritative.

8. Revit Family Audit

Chapter purpose
Describe the family-level audit scope, canonical family evidence, rule classes, finding design, severity-confidence separation, and remediation outputs.

8.1 Canonical family evidence

Family Audit normalizes Revit-family evidence into a canonical family model. Top-level information can include family identity, category, type catalog, parameters, formulas, materials, connectors, nested components, geometry or complexity metrics when available, documentation, source manifest, and organization rules. The purpose of the canonical model is to make family checks independent of the original export format.

The initial export-based implementation should only assess properties that are observable in the supplied evidence. Revit state that has not been extracted is reported as unsupported or not assessed.

8.2 Family rule classes

Identity rules evaluate family name, category, type naming, and duplicate identity conditions. Parameter-governance rules evaluate required parameters, data types, naming, type-versus-instance policy, and shared-parameter expectations. Formula and reference rules evaluate exported dependencies and expected relationships where the data is present. Standards-comparison rules compare observed metadata with explicit customer or organization rule sets.

Complexity indicators and visual-review signals should be treated as calibrated or inferential layers. A screenshot-based observation should not become the sole basis for a critical deterministic finding unless the visual rule has been separately validated.

8.3 Family deliverables

Family Audit produces an executive health summary, family inventory, evidence manifest, categorized findings, remediation checklist, structured findings.json, remediation.csv, and an explicit unsupported or not-assessed section. The output is designed for both human review and downstream automation.

9. BIM Deliverable QA

Chapter purpose
Describe requirement-evidence assessment across BIM execution documentation, schedules, registers, IFC-derived evidence, issue data, and project information deliverables.

9.1 Requirement-Evidence Matrix

BIM QA uses a Requirement-Evidence Matrix as the central assessment artifact. Each requirement receives a stable identifier linked to the source document and location. One or more evidence objects are then associated with the requirement before an assessment state is assigned.

The matrix records requirement_id, normalized requirement text, evidence_refs, assessment, automation_type, confidence, impact, and recommended_action. This structure makes the reasoning path visible and supports re-evaluation when new evidence is supplied.

9.2 Analysis domains

Launch-oriented analysis domains include naming and information-container conventions; required information presence; cross-document identifier, status, and revision consistency; provenance and versioning; defined openBIM or IFC property checks where supported; and propagation of relevant family-data findings into project information quality.

Geometric clash detection, structural verification, MEP performance verification, design suitability, and automatic building-code compliance are outside the initial BIMAP boundary. These domains require specialized deterministic engines, engineering authority, or broader validation than an information-quality service can provide.

9.3 Evidence sufficiency

Accepted file format and sufficient analytical evidence are separate concepts. A technically readable PDF or spreadsheet may still fail to provide the information required for a specific check. The system therefore distinguishes file acceptance from analysis readiness and exposes missing evidence before or during assessment.

10. Combined Audit and Cross-Scope Evidence

Chapter purpose
Explain how family-level and project-level evidence are linked through typed relationships without double-counting or converting correlation into unsupported causation.

10.1 One evidence graph, two audit scopes

Combined Audit stores family and project evidence under one order-level evidence graph or equivalent relational representation. Findings from one scope can reference requirements, schedules, registers, or findings in the other scope without losing their original provenance.

This architecture prevents the combined product from becoming two reports stapled together. Shared identifiers allow the report to show how a family-level data condition relates to an observed project-level information problem.

10.2 Typed relationships

Cross-scope relationships are explicit and typed. causes is reserved for relationships supported strongly enough to establish causal linkage; contributes_to indicates a supporting mechanism without asserting sole causation; corroborates indicates mutually reinforcing evidence; conflicts_with indicates contradictory conditions; depends_on records prerequisite relationships; duplicates captures overlapping findings.

Typed relationships are important because contextual reasoning can otherwise convert correlation or narrative plausibility into unsupported causation. The evidence graph retains both the relationship type and confidence.

10.3 Remediation prioritization

Combined remediation is ordered by dependency, severity, project relevance, confidence, and known effort where sufficient evidence exists. Correcting a shared family parameter defect may resolve multiple downstream schedule inconsistencies; the graph can therefore group related remediation without double-counting the originating defect as multiple independent failures.

11. SLAI Integration and Governed Reasoning

Chapter purpose
Describe the role of SLAI v2.3 as a governed reasoning and orchestration layer operating on normalized evidence and deterministic findings rather than as the public application server.

11.1 SLAI gateway

BIMAP consumes SLAI through a dedicated adapter and job-envelope boundary. The adapter constructs a bounded analysis request from canonical evidence and deterministic findings, invokes only approved SLAI capabilities, and maps governed results back into BIMAP domain objects. The public API does not call agents directly.

This arrangement reflects the service-layer boundary defined for the platform: authentication, orders, uploads, payments, tenant controls, queueing, and customer-visible state remain BIMAP responsibilities; SLAI receives normalized, policy-approved work (Remy3Design, 2026).

11.2 Agent responsibilities

Core SLAI responsibilities can include collaborative workflow coordination, document reading and recovery, project and organization knowledge context, cross-evidence reasoning, remediation planning, quality checks, privacy controls, safety and alignment release policy, evaluation, customer-facing language generation, and runtime observability. Perception is conditional for screenshots or images. Execution is limited to controlled tool or packaging actions. Learning and Adaptive capabilities remain controlled or deferred unless a separate approved feedback policy exists; QNN has no demonstrated product requirement in the launch architecture (Remy3Design, 2026).

BIMAP product rules remain separate from generic SLAI quality thresholds. Internal agent thresholds do not become BIM compliance criteria unless the BIMAP rule system has independently defined and validated that mapping.

11.3 Governed contextual reasoning

Contextual reasoning is used to connect evidence across documents, prioritize findings, identify contradictions, formulate bounded explanations, and synthesize report sections. It does not overwrite deterministic source evidence. Any material inference retains its automation type and confidence and can be routed to review when evidence is insufficient or contradictory.

12. Quality, Security, Privacy, and Human Review

Chapter purpose
Describe infrastructure security, quality gates, privacy controls, release policy, retention, review escalation, and operational safeguards.

12.1 Infrastructure security precedes analysis

File security is an infrastructure responsibility before content reaches the audit engine or SLAI. Layered upload controls include allowlisted extensions, MIME and signature checks where feasible, safe internal filenames, archive limits, storage isolation, authorization, malware scanning, content validation, and resource quotas. These controls follow the defense-in-depth principles described by OWASP for file uploads (OWASP Foundation, n.d.).

Customer objects remain private and are accessed through short-lived signed URLs or equivalent authorized mechanisms. Public object-store paths are not exposed as the customer download interface.

12.2 Privacy and data lifecycle

BIMAP applies purpose limitation, data minimization, storage limitation, confidentiality, and accountability principles to customer evidence. EU data-protection guidance treats these principles and data protection by design and by default as system-level obligations rather than post-processing features (European Commission, 2026; European Data Protection Board, 2020).

Account and order metadata, customer project uploads, normalized evidence, reports, telemetry, and any potential evaluation data are treated as different data classes with separate retention purposes. Customer project content is not assumed to be reusable for model training or product learning.

12.3 Governance gates and human review

Ingress and egress governance are separate. Ingress gates evaluate file security, schema completeness, evidence fitness, and privacy conditions. Egress gates evaluate report structure, contradictory evidence, scope boundaries, unsupported claims, confidence, and release policy. High-severity findings with low confidence, unresolved contradictions, or material recommendations supported only by inference are routed to human review.

Overrides require reason codes and are retained as evaluation evidence. An override does not silently retrain the system.

13. Reporting, Traceability, and Verification

Chapter purpose
Define the human-readable and machine-readable deliverables and explain the evidence-to-finding-to-remediation traceability model.

13.1 Deliverable set

The standard deliverable set includes a professional PDF report, machine-readable findings.json, remediation.csv, evidence_manifest.json, requirement_matrix.csv for BIM QA or Combined Audit, and a packaged audit_bundle.zip. The report is versioned and includes the analysis date, order identifier, rule and schema versions, source evidence hashes, and scope limitations (Remy3Design, 2026).

The PDF is the primary human-readable product, while structured outputs allow findings to enter BIM issue-management, QA, analytics, or later Revit-connected workflows without re-parsing prose.

13.2 Finding traceability

Every finding points to one or more stable evidence objects. The evidence object points to a source file hash and a logical source location. The finding also records the applied rule or requirement, observed and expected values, severity, confidence, remediation, and verification method. A reviewer can therefore move from the report statement back to the evidence and forward to the action required to close it.

Traceability is especially important when automated explanations are used. Generated language must remain bounded by the evidence and rule context and must not introduce unsupported facts.

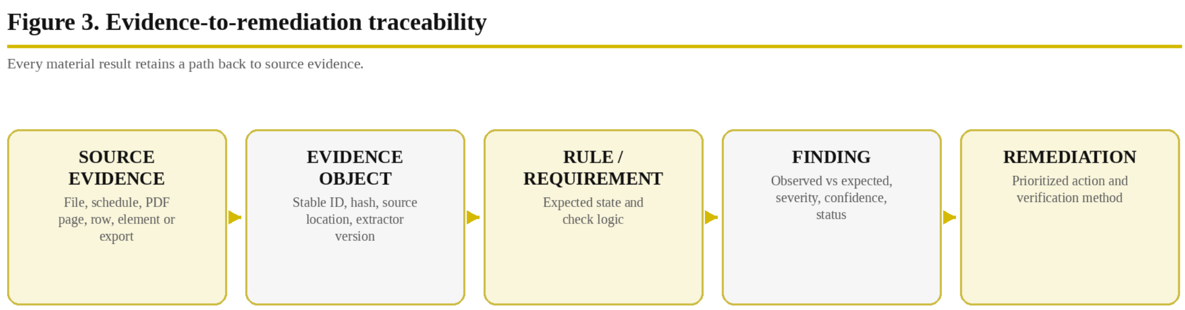

Figure 3. Evidence-to-remediation traceability structure.

13.3 Verification and repeatability

Remediation guidance is paired with a verification method such as re-exporting the family, rerunning a rule, supplying missing evidence, or confirming a corrected schedule or register. This creates a closed quality cycle rather than a one-time narrative assessment.

Versioned schemas and rule sets allow two reports to be distinguished when methodology changes. Reproducibility therefore depends on preserving evidence hashes, rule versions, parser or extractor versions, and report version.

14. Benefits and Advantages

Chapter purpose
Consolidate the technical, operational, governance, interoperability, and workflow benefits created by BIMAP's evidence-first and modular architecture.

14.1 Technical advantages

The separation between ingestion, normalization, deterministic rules, contextual reasoning, governance, and reporting gives each analytical stage a defined responsibility. Deterministic behavior can be benchmarked independently, while contextual functions can be evaluated against evidence-grounding and release criteria. The same canonical evidence objects are reusable across Family Audit, BIM QA, and Combined Audit.

Stable contracts reduce coupling between future Revit extraction and the Python audit engine. A C# exporter can target versioned JSON schemas without importing BIMAP internals, and later cloud automation can produce the same contract.

14.2 Quality-management advantages

Evidence references make findings inspectable rather than opaque. Unknown and manual-review-required states prevent missing information from being misrepresented as certainty. Separating severity from confidence reduces the risk that a severe but weakly supported issue is reported as a certain critical failure.

Typed cross-scope relationships improve combined analysis because family defects can be connected to project information symptoms without automatically multiplying scores or asserting causation. Rule-level validation also permits precision, recall, false-critical rate, and override rate to be measured by check class instead of relying only on overall accuracy.

14.3 Workflow advantages

The asynchronous service model removes the need for a meeting as the default fulfillment mechanism. Customers can submit a bounded evidence package, track controlled status stages, and receive a standardized report and machine-readable outputs. The same order and evidence infrastructure supports all three audit scopes.

Prioritized remediation converts detected defects into an action sequence. Verification methods make the report usable as a quality-control loop: correct, re-export, rerun, and close.

14.4 Governance and trust advantages

The architecture keeps internet-facing controls authoritative for authentication, tenant isolation, storage authorization, payments, and upload security rather than delegating these functions to an agent. Privacy, quality, safety, and evaluation gates remain explicit release stages. Automated findings are labeled by automation type and confidence, which provides a more transparent basis for professional review.

The commercial claim boundary is equally important: BIMAP can communicate evidence-linked quality analysis without representing the service as statutory certification, universal BIM compliance, code checking, or clash detection.

14.5 Interoperability and evolution advantages

Controlled-export support enables an initial implementation without requiring the most complex Revit integration. Versioned contracts then support staged addition of a Revit exporter, IFC/openBIM evidence, and optional Autodesk Platform Services automation. This reduces the risk of coupling first-release logic to a single extraction mechanism.

Machine-readable outputs also create a path for downstream integrations. Findings, requirement matrices, evidence manifests, and remediation actions can be consumed by dashboards, issue systems, internal QA tooling, or future Revit-linked workflows.

14.6 Benefits summary

Benefit	Mechanism	Operational consequence
Traceability	Stable evidence IDs, hashes, rule IDs and source locations	Findings can be reviewed and challenged against source evidence
Reproducibility	Versioned schemas, rules, extractors and reports	Results can be regression-tested and compared across releases
Controlled uncertainty	Unknown states, confidence and manual-review flags	Missing evidence is not converted into false certainty
Cross-scope insight	Typed family-to-project relationships	Remediation can target root information-quality causes
Modularity	Service, audit, SLAI, reporting and storage boundaries	Individual technologies can evolve without rewriting the audit model
Interoperability	JSON/CSV contracts and future IFC/Revit extraction	Results can enter downstream BIM and QA workflows

15. Extensibility: Revit, openBIM, and Cloud Automation

Chapter purpose
Describe the staged path from controlled exports to a Revit add-in, IFC/openBIM expansion, and optional Autodesk Platform Services automation.

15.1 Revit-native exporter

After the evidence contract stabilizes, BIMAP can add a small C# Revit add-in whose responsibility is deterministic extraction rather than reasoning. Autodesk documents the Revit API as suitable for standards enforcement, data extraction, report generation, and application integration (Autodesk, 2026a). The exporter can generate a manifest containing Revit version, document identifier or fingerprint, export timestamp, schema version, exporter version, and source hashes.

Family extraction can include accessible category, type, parameter, formula, material, connector, nested-family, and selected geometry metadata relevant to defined rules. Project extraction can include explicitly supported categories, project parameters, sheets, views, schedules, phases, worksets, design options, warnings, or model summaries where they are part of a validated check.

15.2 IFC and openBIM

IFC provides a vendor-neutral expansion path for project-level QA. buildingSMART describes IFC as an open standard for BIM data exchange, and IFC 4.3 is published as ISO 16739-1:2024 (buildingSMART International, 2024). BIMAP can use IFC-derived evidence for defined property, schema, identity, and information requirements while maintaining the same evidence and finding contracts.

OpenBIM support should be introduced by rule domain rather than by claiming that the presence of an IFC file makes all project requirements machine-checkable.

15.3 Cloud Revit processing

Autodesk Platform Services Revit Automation can later execute Revit add-ins in a cloud-hosted Revit environment for RFA and RVT processing (Autodesk, n.d.). This can remove the customer-side exporter step but introduces cloud cost, Revit-version management, external service dependencies, licensing and operational considerations, and an expanded security boundary.

For that reason, direct RVT or RFA cloud processing is an extension of a validated evidence contract rather than a prerequisite for the first working BIMAP implementation.

16. Validation, Limitations, and Operational Assurance

Chapter purpose
Define validation requirements, benchmark design, explicit exclusions, unknown-state handling, release criteria, and the operational boundary of the service.

16.1 Validation framework

Deterministic rule groups require versioned benchmark sets containing positive, negative, missing, contradictory, and adversarial cases. Expected outcomes should be labeled independently by a BIM-competent reviewer. Performance is measured per rule or rule class using precision, recall, false-critical rate, unknown-state behavior, and regression stability.

Every release that changes rule logic, normalization, extraction, or schemas is regression-tested against the benchmark. During an initial beta, material findings can be manually reviewed and overrides recorded to estimate escape and false-positive rates.

16.2 Explicit limitations

BIMAP cannot assess evidence that has not been supplied or extracted. The platform therefore reports unsupported and not-assessed conditions. The initial service does not provide geometric clash detection, universal BIM compliance certification, building-code compliance, structural design verification, MEP performance verification, or design-suitability approval.

Contextual inference is not equivalent to deterministic evidence. Low-confidence, high-severity results are candidates for review rather than automatic critical claims. Visual observations remain corroborative unless a visual rule has been validated for the specific use case.

16.3 Operational assurance

Operational assurance includes tenant authorization tests, upload and download security review, idempotent payment and queue transitions, auditable retention and deletion tasks, report-withdrawal procedures, dependency and deployment controls, structured logging, and monitoring of queue latency, parser failure, job duration, report-generation failure, and delivery success.

Operational performance and audit accuracy are monitored separately. A fast worker does not imply a correct audit; a correct deterministic rule does not imply that the surrounding service is secure or reliable.

17. Conclusion

Chapter purpose
Summarize BIMAP as a traceable BIM quality-intelligence system and restate the architectural properties that support controlled expansion.

17.1 Summary

BIMAP is structured as an evidence-first BIM quality-intelligence platform that combines deterministic audit logic with bounded, governed contextual reasoning. It supports family-level review, project deliverable QA, and a combined mode that preserves typed relationships between the two scopes. The service is designed around stable evidence identifiers, versioned rules and schemas, explicit unknown states, severity-confidence separation, and deterministic report packaging.

The architecture keeps public service responsibilities outside SLAI, keeps BIM-specific deterministic logic inside the audit engine, and restricts SLAI integration to a controlled gateway. This division supports validation, security, portability, and future integration with Revit-native extraction, IFC/openBIM evidence, and cloud model automation.

The central product outcome is a traceable chain from source evidence to rule or requirement, from rule result to finding, and from finding to prioritized remediation and verification. This structure provides the basis for repeatable BIM quality review without overstating the certainty or professional authority of automated analysis.

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References are presented in APA style as closely as practical for a technical whitepaper. Online technical sources and standards should be rechecked when they are used as contractual or compliance authorities.

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